Technical Supplement for SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse HER Actor-Critic

Anonymous submission

Abstract

This supplement provides complete experimental settings, seed-level primary results, sensitivity and scope controls, support-signal diagnostic details, runtime measurements, and an index of the anonymous code-and-data artifacts. It uses the terminology and method definitions of the main paper.

Working-draft completion status. Two result slots are intentionally marked *pending*: FPP DDPG Actor+Critic and DrawerOpen DDPG fixed-target diagnostics. No value is inferred from an incomplete run or from the earlier live-target diagnostic. These markers and this paragraph must be removed only after complete, validated artifacts have replaced both slots.

1 Experimental and Reproducibility Details

1.1 Tasks, Cohorts, and Shared Training Settings

Tables 1–3 give the settings shared within each matched cohort. The primary comparison contains seven task–backbone cohorts and uses paired seeds 1–5. Drawer SAC is not a primary cohort because a complete matched HER–SABER pair was not collected.

HER convention. For an episode with transitions indexed by $t = 0, \ldots, T-1$, a transition selected for relabeling samples

$$u \sim \mathrm{Unif}\{t+1, \ldots, T-1\}, \qquad \widetilde{g} = m(s_{u+1}).$$

The original desired goal is retained when the eligible set is empty or when relabeling is not selected. Reward is recomputed using $\widetilde{g}$. Consequently, the nominal relabeling probability is 0.8, while the realized fraction may be slightly lower near an episode boundary. Matched HER–SABER cohorts execute the same replay and relabeling code.

Terminal convention. A successfully relabeled transition is treated as nonterminal for bootstrap purposes. Time-limit transitions also bootstrap. This convention is identical across methods within each cohort.

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Task	Horizon	Budget	Backbones	Exact primary grid
FetchPickAndPlace-v4	50	1M	DDPG, TD3, SAC	10k (DDPG/TD3), 50k (SAC)
DrawerOpen	200	2M	DDPG, TD3	10k
DrawerClose	200	2M	DDPG, TD3	10k

Table 1: Primary task and cohort definitions. Every method within a task–backbone cohort uses the same environment-step budget and exact evaluation grid. Primary comparisons use paired seeds 1–5.

Setting	Value or convention
HER	Future-goal convention in the main paper; nominal $p_{\mathrm{HER}} = 0.8$
Replay	Uniform for HER and SABER; capacity 300k; minibatch size 256
Networks	Actor and critic MLPs with two hidden layers of 256 units
Optimization	Adam, actor and critic learning rates 3×10^{-4}
Discount and targets	$\gamma = 0.98$, Polyak coefficient $\tau = 0.005$
Update schedule	UTD 1; learning begins after 5k steps; random actions for first 10k steps
Termination	Successful relabeled transitions are nonterminal; time limits bootstrap
Evaluation	Deterministic policy, 20 episodes per checkpoint
Statistics	Seed mean and sample SD; paired percentile-bootstrap CI, 10,000 resamples, RNG seed 2027
Curve processing	Exact common checkpoints only; no smoothing, interpolation, extrapolation, or filling

Table 2: Shared training, evaluation, and statistical settings. Matched HER–SABER runs use identical replay and relabeling code.

Backbone	Critic	Exploration or entropy	Backbone-specific settings
DDPG	One critic	Gaussian action noise, SD 0.2	Deterministic actor
TD3	Twin critics	Gaussian action noise, SD 0.2	Target noise 0.2, clip 0.5, policy delay 2
SAC	Twin critics	Automatic entropy tuning	Initial $\alpha = 0.2$, α LR 3×10^{-4}, target entropy -4

Table 3: Backbone-specific settings. SABER preserves each backbone's target construction and actor-update rule.

1.2 Support Context and Retrieval

The replay index returns at most N_{scan} slots associated with a coarse retrieval key in implementation-defined index order. Those candidates are filtered by exact component-wise equality of the full context. The current implementation does not guarantee temporal recency after circular-buffer wraparound; no claim relies on candidates being the most recent transitions. The sampled transition remains eligible to appear in its own local replay set.

For the Drawer phase definitions, let d_h be hand–handle distance and d_g be handle–goal distance. The terminal phase is open or closed when $d_g \leq 0.05$; the opening or closing phase requires $d_h \leq 0.08$ and $0.05 < d_g \leq 0.12$; contact requires $d_h \leq 0.08$ and $d_g > 0.12$; remaining samples are approach. Phases use the relabeled training goal.

Task family	Full context components	Coarse retrieval key
FetchPickAndPlace	Phase, hand–object distance, object–goal distance, object height, gripper gap, horizon	Hand–object distance, object height, gripper gap, horizon
DrawerOpen/Close	Phase, hand–handle distance, handle–goal distance, handle y, gripper state, horizon	Hand–handle distance, handle y, gripper state, horizon

Table 4: Task-specific support contexts. Candidate retrieval uses the coarse key, after which exact component-wise equality on the full discretized context defines the local replay set.

Component	FetchPickAndPlace thresholds	Drawer thresholds
Phase	approach, grasp, lift, place	approach, contact, opening/closing, open/closed
Hand distance	$.03, .05, .08, .12, .20, .40$	$.03, .05, .08, .12, .20, .40$
Goal distance	$.05, .10, .20, .40, .70$	$.03, .05, .08, .12, .20, .40, .70$
Task coordinate	object z: $.43, .48, .55, .65, .75$	handle y: $-.05, 0, .05, .10, .15, .25$
Gripper	gap: $.02, .05, .08$	state: $-.50, 0, .50$
Remaining horizon	$10, 25, 50, 100$	$10, 25, 50, 100, 200, 500$

Table 5: Discretization boundaries used to construct support contexts. Values are applied component-wise in environment units. The main method normalizes each action dimension to its action bounds and uses $K = 5$ bins per dimension, $N_{\text{scan}} = 512$, $N_{\text{local}} = 4$, and $\lambda_S = 1$.

1.3 Comparison Methods

HGR. The implementation prioritizes an episode and then a transition–future-goal pair. The two importance factors are multiplied and normalized on relabeled samples; the original-goal branch retains coefficient one. The episode and pair priority exponents are both 0.6.

PER. Replay transitions are sampled by TD-error priority with exponent 0.6. The importance exponent increases linearly from 0.4 to 1.0 over the training budget. PER changes realized minibatch membership, unlike SABER and the TD-error critic-loss weighting control.

TD-error critic-loss weighting. This control preserves uniform replay and changes only normalized per-sample critic coefficients. DDPG uses its single absolute TD error. SAC uses $\min(|\delta_{i1}|, |\delta_{i2}|)$ as the signal and applies the resulting coefficient to both critic losses. TD3 results are not reported because the stored control used independent target-policy smoothing noise for weight construction and optimized residuals, preventing the intended matched-signal interpretation.

Method	Selection changed	Loss coefficient	Correction
HER	None beyond uniform replay and HER relabeling	Uniform	None
SABER	None	Current-policy local-support weight	Unit minibatch mean
HGR	Episode and transition–future-goal pair	HGR importance weight	Two-stage IS
PER	Replay transition	PER importance weight	Annealed IS
TD-error weighting	None	TD-error-derived weight	Unit minibatch mean

Table 6: Operational intervention points. HGR uses priority exponents $\alpha_{\text{episode}} = \alpha_{\text{pair}} = 0.6$, while PER uses $\alpha = 0.6$. Both anneal β linearly from 0.4 to 1.0 over critic-update progress and use priority offset 10^{-6}. TD-error weighting retains uniform replay and is therefore the closest allocation-stage control.

1.4 Metrics and Statistical Analysis

Let $q_j \in [0, 1]$ be deterministic evaluation success at exact shared checkpoint t_j. Normalized success AUC is

$$\text{AUC} = \frac{1}{t_m - t_1} \sum_{j=1}^{m-1} \frac{q_j + q_{j+1}}{2} (t_{j+1} - t_j).$$

Final-quarter AUC applies the same calculation to 750k–1M for FPP and 1.5M–2M for Drawer tasks. Final success is the value at the final checkpoint. Hit90 is the first exact checkpoint with $q_j \geq 0.9$; its median is computed only over seeds that reach the threshold, so reached-seed coverage is always reported.

For paired comparisons, each seed contributes one method difference. The reported 95% intervals are percentile intervals from 10,000 paired bootstrap resamples with pseudo-random seed 2027. Sample standard deviations use $n - 1$ in the denominator. No curve is smoothed, interpolated, extrapolated, or forward-filled.

2 Primary Comparison Details

2.1 Drawer Learning Curves

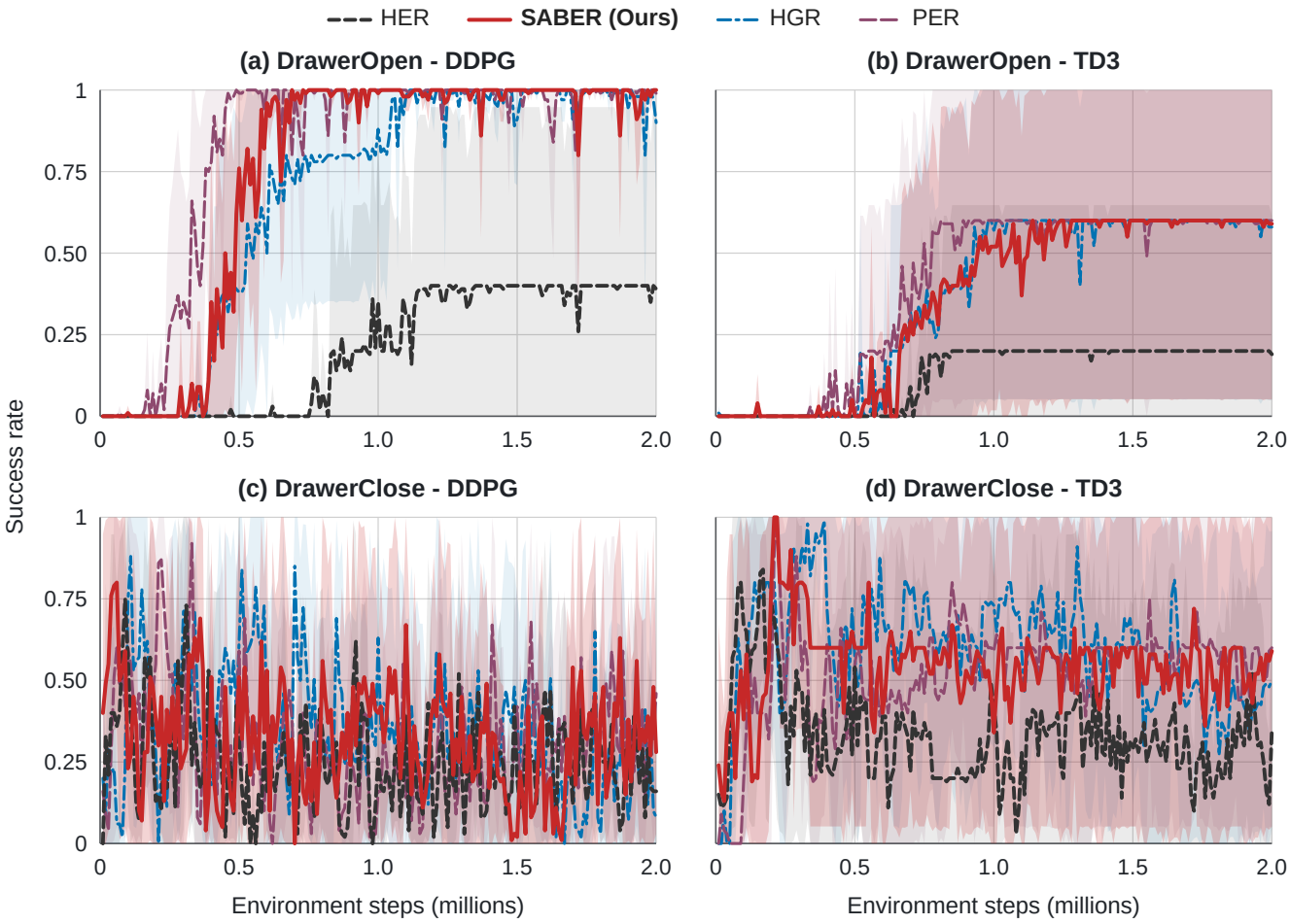


Figure 1: DrawerOpen and DrawerClose learning curves for DDPG and TD3. Lines are seed means and bands are sample standard deviations over paired seeds 1–5 on the exact 10k grid. No smoothing or interpolation is applied.

2.2 Time to Threshold

Hit90 is a threshold-crossing summary rather than a stability measure. Coverage is especially important when only a subset of seeds reaches 0.9.

Task	Backbone	Method	Median Hit90	Reached seeds
FetchPickAndPlace	DDPG	SABER	540k	5/5
		HER	600k	5/5
		HGR	930k	2/5
		PER	640k	4/5
FetchPickAndPlace	TD3	SABER	840k	5/5
		HER	980k	3/5
		HGR	630k	1/5
		PER	1.00M	1/5
FetchPickAndPlace	SAC	SABER	750k	5/5
		HER	800k	3/5
		HGR	800k	2/5
		PER	550k	5/5
DrawerOpen	DDPG	SABER	500k	5/5
		HER	960k	2/5
		HGR	530k	5/5
		PER	330k	5/5
DrawerOpen	TD3	SABER	780k	3/5
		HER	740k	1/5
		HGR	760k	3/5
		PER	720k	3/5
DrawerClose	DDPG	SABER	20k	5/5
		HER	70k	5/5
		HGR	100k	5/5
		PER	60k	5/5
DrawerClose	TD3	SABER	80k	5/5
		HER	70k	5/5
		HGR	80k	5/5
		PER	120k	5/5

Table 7: Hit90 for the primary comparisons. Hit90 is the first exact-grid checkpoint at which deterministic evaluation success reaches at least 0.9. Medians use only reached seeds, so reached-seed coverage is reported separately.

2.3 FPP TD3 Confirmatory and HER-Indexing Sensitivity Results

The ten-seed cohort extends the frozen primary FPP TD3 comparison. The inclusive-future sensitivity also admits $m(s_{t+1})$, while leaving the matched HER–SABER difference otherwise unchanged. Its positive observed mean direction does not constitute an equivalence test between HER indexing conventions.

Seed	HER AUC	SABER AUC	Difference
1	0.3465	0.4301	$+0.0836$
2	0.3727	0.4568	$+0.0841$
3	0.3942	0.4960	$+0.1018$
4	0.4109	0.4066	-0.0043
5	0.3338	0.4225	$+0.0886$
6	0.4301	0.5207	$+0.0907$
7	0.3995	0.4614	$+0.0619$
8	0.3068	0.3543	$+0.0475$
9	0.4106	0.4114	$+0.0008$
10	0.4210	0.3331	-0.0879
Mean	0.3826	0.4293	$+0.0467$

Table 8: Frozen FPP TD3 confirmatory extension over ten paired seeds. AUC is computed on the exact 10k–1M evaluation grid without smoothing, interpolation, or filling. The paired mean difference is $+0.0467$ (10,000-resample paired-bootstrap 95% CI $[0.0085, 0.0783]$); 8/10 paired differences are positive. The exact two-sided sign-flip test over all 1,024 assignments gives $p = 0.0527$.

Seed	HER AUC	SABER AUC	Difference
1	0.3005	0.4576	$+0.1571$
2	0.2674	0.2669	-0.0005
3	0.3467	0.5179	$+0.1712$
4	0.3487	0.4003	$+0.0515$
5	0.3669	0.3247	-0.0422
Mean	0.3261	0.3935	$+0.0674$

Table 9: FPP TD3 sensitivity to inclusive future-goal indexing, which also admits $m(s_{t+1})$. AUC uses the exact 10k–1M evaluation grid. The paired mean difference is $+0.0674$ (10,000-resample paired-bootstrap 95% CI $[-0.0068, 0.1416]$), with 3/5 positive pairs and an exact two-sided sign-flip test of $p = 0.25$. The final-quarter AUC difference is $+0.1854$ (95% CI $[-0.0212, 0.3920]$), and the final-success difference is $+0.2200$ (95% CI $[-0.2000, 0.5200]$).

3 Additional Controls and Sensitivities

3.1 Allocation Signal

Across the validated DDPG and SAC cohorts in Table 10, SABER has higher observed mean AUC than TD-error critic-loss weighting. These controls isolate the weighting signal more closely than PER because both retain uniform replay.

Task	Backbone	Method	Final	AUC	Hit90
FetchPickAndPlace	DDPG	SABER	0.960 ± 0.042	0.609 ± 0.038	540k (5/5)
FetchPickAndPlace	DDPG	TD-error	0.930 ± 0.104	0.568 ± 0.069	630k (5/5)
DrawerOpen	DDPG	SABER	1.000 ± 0.000	0.754 ± 0.044	500k (5/5)
DrawerOpen	DDPG	TD-error	0.800 ± 0.447	0.715 ± 0.090	610k (5/5)
DrawerClose	DDPG	SABER	0.280 ± 0.251	0.324 ± 0.082	20k (5/5)
DrawerClose	DDPG	TD-error	0.400 ± 0.166	0.313 ± 0.098	130k (5/5)
FetchPickAndPlace	SAC	SABER	0.990 ± 0.022	0.583 ± 0.030	750k (5/5)
FetchPickAndPlace	SAC	TD-error	0.990 ± 0.022	0.544 ± 0.068	800k (5/5)

Table 10: Validated TD-error critic-loss weighting controls. Replay sampling remains uniform, and only critic-loss coefficients are changed. Values are mean $\pm$ sample standard deviation over paired seeds 1–5. SABER rows reproduce the matched main-method values; TD-error rows use TD-error critic-loss weighting. For SAC, the priority signal is $\min(|\delta_{i1}|, |\delta_{i2}|)$, and the resulting normalized coefficient is applied to both critic losses.

3.2 Action-Bin Resolution

Both $K = 3$ and $K = 7$ lower FPP TD3 mean AUC relative to the Main $K = 5$ setting in this paired sensitivity. The accompanying support rates show that bin resolution also changes activation frequency; this experiment does not isolate bin geometry from support density.

Action bins	Support-positive rate	AUC	FQ-AUC	Final	ΔAUC vs. $K = 5$
K=3	0.396 ± 0.012	0.353 ± 0.048	0.603 ± 0.121	0.750 ± 0.265	$-0.089[-0.109, -0.070]$
K=5 (Main)	0.219 ± 0.009	0.442 ± 0.035	0.804 ± 0.103	0.970 ± 0.045	$--$
K=7	0.148 ± 0.010	0.361 ± 0.047	0.616 ± 0.151	0.690 ± 0.263	$-0.081[-0.093, -0.067]$

Table 11: FPP TD3 action-bin sensitivity over paired seeds 1–5. Values are mean $\pm$ sample standard deviation; FQ-AUC is the final-quarter normalized success AUC. The final column reports variant-minus-Main paired AUC differences with 10,000-resample percentile-bootstrap 95% confidence intervals.

3.3 Actor and Critic Weight Scope

Scope variants hold the support-query rule, support hyperparameters, and minibatch normalization fixed. Critic-Only applies Ω_i only to Bellman losses; Actor-Only applies it to per-sample actor terms; Actor+Critic applies it to both. Because each scope induces its own training trajectory, realized support indicators are not assumed to be identical across runs.

Task	Scope	AUC	FQ-AUC	Final	Hit90	ΔAUC (Main–scope)
FPP	Critic-Only (Main)	0.609 ± 0.038	0.961 ± 0.014	0.960 ± 0.042	5/5 (0.540M)	$--$
	Actor-Only	0.589 ± 0.032	0.952 ± 0.016	0.960 ± 0.022	5/5 (0.600M)	$+0.020[-0.024, 0.063]$
	Actor+Critic			*pending*		
DrawerOpen	Critic-Only (Main)	0.754 ± 0.044	0.987 ± 0.018	1.000 ± 0.000	5/5 (0.500M)	$--$
	Actor-Only	0.612 ± 0.343	0.760 ± 0.433	0.800 ± 0.447	4/5 (0.470M)	$+0.141[-0.051, 0.478]$
	Actor+Critic	0.728 ± 0.084	0.992 ± 0.008	0.990 ± 0.022	5/5 (0.540M)	$+0.026[-0.048, 0.091]$
DrawerClose	Critic-Only (Main)	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	$--$
	Actor-Only	0.307 ± 0.039	0.299 ± 0.061	0.280 ± 0.299	5/5 (0.040M)	$+0.017[-0.026, 0.078]$
	Actor+Critic	0.298 ± 0.071	0.241 ± 0.119	0.440 ± 0.370	5/5 (0.050M)	$+0.026[-0.043, 0.091]$

Table 12: DDPG actor-scope ablations over paired seeds 1–5. Values are mean $\pm$ sample standard deviation. Hit90 reports the number of reaching seeds and the reached-seed median step. The final column gives Main-minus-scope paired AUC differences with 95% bootstrap confidence intervals. Pending cells are left unreported until a complete five-seed cohort is available.

3.4 Secondary Replay and Update-Budget References

The no-HER AER comparisons test a different setting from the paper's post-HER question. The high-replay-ratio and REDQ-HER results demonstrate that additional updates and a larger critic ensemble can be stronger on FPP SAC, but they change UTD or architecture and therefore do not replace the matched UTD-1 comparison.

Backbone	No-HER replay	AUC	Final
DDPG	ER	0.148 ± 0.039	0.460 ± 0.188
	Annealing ER	0.129 ± 0.111	0.340 ± 0.363
	AER	0.044 ± 0.002	0.000 ± 0.000
TD3	ER	0.039 ± 0.002	0.000 ± 0.000
	Annealing ER	0.039 ± 0.002	0.000 ± 0.000
	AER	0.039 ± 0.002	0.000 ± 0.000
SAC	ER	0.064 ± 0.042	0.130 ± 0.199
	Annealing ER	0.041 ± 0.002	0.000 ± 0.000
	AER	0.040 ± 0.003	0.000 ± 0.000

Table 13: FPP no-HER replay references over seeds 1–5. These runs omit hindsight relabeling and are context references rather than matched post-HER mechanism controls.

Method	UTD	Critic configuration	AUC	Final	Hit90
HER	1	2 critics	0.471 ± 0.079	0.790 ± 0.288	3/5 (0.80M)
SABER	1	2 critics	0.583 ± 0.030	0.990 ± 0.022	5/5 (0.75M)
HER high-RR	4	2 critics	0.768 ± 0.097	0.960 ± 0.042	5/5 (0.45M)
REDQ-HER TQB	4	10 critics, 2 sampled	0.867 ± 0.052	1.000 ± 0.000	5/5 (0.30M)

Table 14: FPP SAC scaling references on the exact shared 50k–1M grid. High-replay-ratio HER and REDQ-HER change the update budget, and REDQ-HER also changes critic architecture; they are not matched post-HER loss-allocation controls.

4 Support-Signal Diagnostic Details

4.1 Instrumentation and Aggregation

Diagnostic logging introduces no gradient term and does not change replay sampling, objectives, or parameter-update order. FPP logs one full minibatch every 10k environment steps for DDPG, TD3, and SAC seeds 1–5: 100 batches per run, 500 per backbone, and 1,500 total. Drawer diagnostics use a 200-step cadence from 5k through 2M, producing 9,976 persisted snapshots per seed. Snapshot quantities are aggregated within seed before cohort statistics are computed.

For each persisted minibatch, the diagnostic stores the pre-update absolute Bellman residual, support membership, and the residual after one critic update. The fixed-target residual is

$$\Delta|\delta_i| = \left|Q_\phi^{\mathrm{pre}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right| - \left|Q_\phi^{\mathrm{post}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right|.$$

Thus the target and minibatch are held fixed while the critic parameters change. Live-target quantities, where available, are retained only for audit and are not mixed with the fixed-target statistic.

4.2 Cohort-Level Results

Top-10% enrichment is the support-positive rate among the exact $\lceil 0.1|B| \rceil$ samples with largest pre-update absolute residual, divided by the full-minibatch support-positive rate. Stable descending order breaks ties, so the selected set has exactly the intended size. The modest Pearson correlations and enrichment above one show that support and TD error are related but not identical rankings.

Cohort	Seeds	Support rate	Pearson r	Top-10% enrich.	Pre-TD $+/-$	Fixed $\Delta\|\delta\|$ $+/-$
FPP DDPG	1–5	0.367 ± 0.011	0.116	$1.409\times$	$1.42\times$	$2.39\times$
FPP TD3	1–5	0.219 ± 0.009	0.128	$1.670\times$	$1.62\times$	$3.30\times$
FPP SAC	1–5	0.197 ± 0.013	0.143	$1.828\times$	$1.74\times$	$3.94\times$
DrawerOpen DDPG	1–5			*pending fixed-target result*		
DrawerOpen DDPG	6–10 extension			*pending fixed-target result*		
DrawerOpen DDPG	1–10 pooled			*pending fixed-target result*		
DrawerClose DDPG	1–5	0.298 ± 0.018	0.104	$1.534\times$	$1.42\times$	$2.94\times$
DrawerClose DDPG	6–10 extension	0.299 ± 0.007	0.118	$1.602\times$	$1.48\times$	$4.23\times$
DrawerClose DDPG	1–10 pooled	0.298 ± 0.013	0.111	$1.568\times$	$1.45\times$	$3.48\times$

Table 15: Support-signal diagnostics. Statistics are first aggregated within seed, after which seeds are independent units. Support rates are mean $\pm$ sample SD. The last two columns divide support-positive by support-negative cohort means. Fixed $\Delta|\delta|$ evaluates the same minibatch after one critic step while retaining the pre-update Bellman target. DrawerOpen values are intentionally unreported until its fixed-target diagnostic rerun is complete.

4.3 Phase-Conditioned Allocation

For DrawerClose, both the primary seeds and the extension show higher support activation in approach and contact than in the dominant closed phase. This phase analysis describes where the indicator activates; it is not an independent causal test of long-horizon performance.

Cohort	Seeds	Phase	Batch fraction	Support rate	Pre-update $\lvert\delta\rvert$	Fixed $\Delta\lvert\delta\rvert$
DrawerOpen DDPG	1–5			*pending fixed-target result*		
DrawerOpen DDPG	6–10 extension			*pending fixed-target result*		
DrawerClose DDPG	1–5	Approach	0.218 ± 0.002	0.549 ± 0.032	0.173 ± 0.010	0.0081 ± 0.0018
		Contact	0.016 ± 0.003	0.710 ± 0.026	0.262 ± 0.011	0.0147 ± 0.0018
		Closing	0.019 ± 0.004	0.425 ± 0.034	0.381 ± 0.066	0.0232 ± 0.0067
		Closed	0.747 ± 0.004	0.212 ± 0.017	0.062 ± 0.009	0.0011 ± 0.0014
DrawerClose DDPG	6–10 extension	Approach	0.222 ± 0.003	0.569 ± 0.004	0.187 ± 0.022	0.0081 ± 0.0007
		Contact	0.018 ± 0.003	0.727 ± 0.016	0.282 ± 0.007	0.0169 ± 0.0019
		Closing	0.017 ± 0.002	0.438 ± 0.028	0.401 ± 0.053	0.0248 ± 0.0033
		Closed	0.743 ± 0.007	0.204 ± 0.010	0.069 ± 0.011	0.0006 ± 0.0006

Table 16: Drawer phase-conditioned diagnostics. Phases are defined from the relabeled training goal. Values are seed means $\pm$ sample SD. The seeds 6–10 rows are a separate diagnostic extension; pooled ten-seed and seed-level values are included in the accompanying artifact.

5 Runtime and Resource Measurements

The dedicated profile uses one hardware/software stack, one seed, and the 20k–100k interval. It supports attribution of implementation overhead but not a claim about asymptotic complexity or end-to-end learning efficiency. The broader HGR comparison uses recorded training throughput and lacks uniform hardware metadata; only within-cohort ratios are interpreted. Peak memory was not recorded and is therefore reported as not measured rather than estimated.

Component	HER ms/step	SABER ms/step	Difference	SABER share
Interaction	1.000	1.071	+0.070	4.4%
Replay + HER	1.925	2.651	+0.726	11.0%
Batch preparation	0.096	0.108	+0.012	0.4%
Support and weighting	0.000	15.028	+15.028	62.3%
Critic update	2.945	2.832	−0.113	11.7%
Actor + target update	1.384	1.292	−0.091	5.4%
Evaluation/logging amortized	0.080	0.085	+0.005	0.4%
Total training loop	8.063	24.133	+16.070	100.0%

Table 17: Hardware-specific FPP TD3 implementation profile over the 20k–100k window of seed 1. HER and SABER achieve 124.0 and 41.4 environment steps/s excluding evaluation, respectively. The profile identifies current support retrieval as the dominant overhead; it is not a learning-performance claim and is not extrapolated to 1M steps.

Task	Backbone	HGR	SABER	Ratio
FPP	DDPG	16.93	41.72	2.46×
FPP	TD3	12.94	41.25	3.19×
FPP	SAC	8.39	29.10	3.47×
DrawerOpen	DDPG	13.24	23.33	1.76×
DrawerOpen	TD3	11.65	21.57	1.85×
DrawerClose	DDPG	13.22	23.96	1.81×
DrawerClose	TD3	11.81	22.49	1.90×

Table 18: Recorded environment steps/s for HGR and SABER. Values are descriptive means over seeds 1–5 within each cohort. Hardware identity is not uniformly recorded across cohorts, so this supports only within-cohort implementation comparisons and no cross-cohort hardware ranking.

6 Artifact and Provenance Index

Generated analysis directories contain compact CSV summaries, a repository-relative source manifest, and a self-excluding artifact manifest. Canonical experiment artifacts retain configuration, environment snapshot, source provenance, and validation records where available. Raw result files are referenced rather than duplicated in the derived summaries.

Content	Artifact key	Status
Primary seed metrics and curves	primary_seed_results	Complete
Action-bin sensitivity	action_bin_sensitivity	Complete
DDPG actor-scope analysis	actor_scope_ddpg	FPP A+C pending
Derived Technical Supplement summaries	technical_supplement	Open fixed-target pending
FPP diagnostics	fpp_diagnostic	Complete
DrawerClose diagnostics	drawerclose_diagnostic	Complete
DrawerOpen fixed-target diagnostics	draweropen_fixed_target	Pending
FPP TD3 runtime profile	fpp_td3_runtime	Complete

Table 19: Artifact index. Keys resolve to repository-relative package paths in artifact_index.csv. Canonical artifacts include manifests that bind source files by repository-relative path and SHA256.

A Primary Seed-Level Results

The following tables report every primary seed value used in the aggregate comparison. AUC and final-quarter AUC use each cohort's exact common grid.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.640152	0.956000	0.900	0.510M
	2	0.537374	0.934000	0.950	0.600M
	3	0.578788	0.964000	0.950	0.610M
	4	0.325253	0.731000	0.900	0.920M
	5	0.612374	0.967000	1.000	0.570M
SABER	1	0.656061	0.939000	0.950	0.480M
	2	0.615152	0.972000	0.950	0.540M
	3	0.564899	0.959000	1.000	0.630M
	4	0.631818	0.976000	1.000	0.480M
	5	0.577525	0.960000	0.900	0.610M
HGR	1	0.441667	0.765000	1.000	0.870M
	2	0.076515	0.065000	0.000	–
	3	0.374495	0.482000	0.500	–
	4	0.417677	0.587000	0.700	0.990M
	5	0.380556	0.506000	0.500	–
PER	1	0.484091	0.919000	1.000	0.770M
	2	0.520455	0.953000	1.000	0.730M
	3	0.604293	0.946000	0.900	0.550M
	4	0.643434	0.964000	1.000	0.460M
	5	0.049242	0.046000	0.000	–

Table 20: Seed-level primary results for FPP with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.346465	0.523000	0.550	–
	2	0.372727	0.635000	0.950	1.000M
	3	0.394192	0.662000	1.000	0.980M
	4	0.410859	0.686000	0.950	0.970M
	5	0.333838	0.530000	0.750	–
SABER	1	0.430051	0.846000	1.000	0.840M
	2	0.456818	0.856000	1.000	0.860M
	3	0.495960	0.914000	0.950	0.630M
	4	0.406566	0.652000	0.900	0.630M
	5	0.422475	0.752000	1.000	0.910M
HGR	1	0.345202	0.515000	0.500	–
	2	0.396212	0.593000	0.800	–
	3	0.407576	0.581000	0.850	–
	4	0.326515	0.489000	0.450	0.630M
	5	0.361111	0.493000	0.450	–
PER	1	0.345455	0.515000	0.550	–
	2	0.408586	0.653000	0.950	1.000M
	3	0.394192	0.628000	0.800	–
	4	0.369192	0.526000	0.500	–
	5	0.327020	0.488000	0.500	–

Table 21: Seed-level primary results for FPP with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.559211	0.985000	1.000	0.800M
	2	0.469737	0.760000	1.000	0.950M
	3	0.406579	0.500000	0.500	–
	4	0.540789	0.965000	1.000	0.800M
	5	0.378947	0.465000	0.450	–
SABER	1	0.636842	0.995000	0.950	0.650M
	2	0.567105	0.980000	1.000	0.800M
	3	0.573684	0.980000	1.000	0.750M
	4	0.575000	1.000000	1.000	0.750M
	5	0.564474	0.975000	1.000	0.800M
HGR	1	0.451316	0.545000	0.800	–
	2	0.596053	0.980000	1.000	0.700M
	3	0.506579	0.800000	1.000	0.900M
	4	0.450000	0.505000	0.500	–
	5	0.414474	0.490000	0.500	–
PER	1	0.755263	0.970000	1.000	0.450M
	2	0.635526	0.985000	1.000	0.550M
	3	0.706579	0.990000	1.000	0.550M
	4	0.690789	1.000000	1.000	0.550M
	5	0.660526	0.985000	0.950	0.600M

Table 22: Seed-level primary results for FPP with SAC. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.460302	0.970000	1.000	1.090M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000000	0.000000	0.000	–
	5	0.576256	0.993500	0.950	0.830M
SABER	1	0.757286	1.000000	1.000	0.510M
	2	0.786935	0.977000	1.000	0.420M
	3	0.705025	1.000000	1.000	0.580M
	4	0.713065	0.960000	1.000	0.500M
	5	0.806030	1.000000	1.000	0.400M
HGR	1	0.799749	0.995000	0.500	0.410M
	2	0.704523	0.990000	1.000	0.530M
	3	0.692211	0.993000	1.000	0.610M
	4	0.794975	0.998000	1.000	0.430M
	5	0.449497	0.934500	1.000	1.060M
PER	1	0.859799	1.000000	1.000	0.330M
	2	0.877387	0.998000	1.000	0.270M
	3	0.814070	0.969000	1.000	0.330M
	4	0.809171	0.951500	0.950	0.390M
	5	0.765075	0.990000	1.000	0.410M

Table 23: Seed-level primary results for DrawerOpen with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.632286	0.999500	0.950	0.740M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000251	0.000000	0.000	–
	5	0.000000	0.000000	0.000	–
SABER	1	0.486558	0.996500	0.950	1.070M
	2	0.000000	0.000000	0.000	–
	3	0.000503	0.000000	0.000	–
	4	0.689950	0.997000	1.000	0.560M
	5	0.632161	0.997000	1.000	0.780M
HGR	1	0.000000	0.000000	0.000	–
	2	0.692462	0.992000	1.000	0.520M
	3	0.531910	0.997000	0.900	0.930M
	4	0.000000	0.000000	0.000	–
	5	0.628266	0.996000	1.000	0.760M
PER	1	0.632412	0.986000	1.000	0.780M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.755528	0.999000	1.000	0.520M
	5	0.644221	0.994000	1.000	0.720M

Table 24: Seed-level primary results for DrawerOpen with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.281658	0.337500	0.300	0.090M
	2	0.268844	0.315500	0.100	0.150M
	3	0.203518	0.193000	0.400	0.040M
	4	0.249497	0.289000	0.000	0.070M
	5	0.251508	0.116000	0.000	0.020M
SABER	1	0.287814	0.165500	0.650	0.010M
	2	0.337437	0.417000	0.200	0.060M
	3	0.284045	0.306500	0.150	0.050M
	4	0.461307	0.300500	0.000	0.020M
	5	0.251508	0.197500	0.400	0.010M
HGR	1	0.672613	0.218000	0.000	0.100M
	2	0.323869	0.236500	0.000	0.090M
	3	0.237563	0.154500	0.150	0.120M
	4	0.238568	0.219500	0.050	0.100M
	5	0.273367	0.299000	0.200	0.010M
PER	1	0.288568	0.272000	0.550	0.100M
	2	0.244975	0.273500	0.000	0.180M
	3	0.297990	0.392000	1.000	0.040M
	4	0.331281	0.356500	0.350	0.050M
	5	0.376884	0.404000	0.400	0.060M

Table 25: Seed-level primary results for DrawerClose with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.450628	0.455000	0.850	0.080M
	2	0.058291	0.002000	0.000	0.120M
	3	0.886181	0.847000	0.900	0.050M
	4	0.097236	0.017000	0.000	0.060M
	5	0.151382	0.098000	0.000	0.070M
SABER	1	0.080653	0.020000	0.000	0.080M
	2	0.908291	0.978000	1.000	0.090M
	3	0.739950	0.761000	1.000	0.200M
	4	0.908920	0.909500	0.950	0.010M
	5	0.106030	0.000000	0.000	0.050M
HGR	1	0.923618	0.886000	0.400	0.080M
	2	0.686307	0.808500	0.950	0.060M
	3	0.782161	0.428000	1.000	0.130M
	4	0.477010	0.115000	0.150	0.060M
	5	0.138945	0.063000	0.000	0.260M
PER	1	0.111055	0.050000	0.000	0.220M
	2	0.845477	0.987000	1.000	0.190M
	3	0.824874	0.891000	1.000	0.120M
	4	0.112060	0.002000	0.000	0.120M
	5	0.725377	0.983000	1.000	0.100M

Table 26: Seed-level primary results for DrawerClose with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.